# Emilia Ma

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## Education

**University of Toronto** Sept 2024 – Expected May 2029
B.Sc. in Computer Science — Dean's List Scholar (2025, 2026) GPA: 3.76/4.0
- Coursework: Data Structures & Algorithms, Software Design, Human Computer Interaction, Systems Programming, Computer Systems & Digital Logic, Theory of Computation, Probability & Statistics

## Experience

**Ontario Public Service | Backend Dev/Junior Technical Analyst** Sept 2026 – Present
- Restored the order orchestrator's Bruno backend test suite from 19 failing scripts to 0 across order creation, driver licence, cross domain, and payment workflows.
- Implemented test-runner logic that queries a Spring Boot Actuator endpoint for configuration resolved from active profiles and runs transaction reversal only for confirmed real downstream calls, skipping mocked and unknown cases to prevent false failures from mock-generated transaction IDs.
- Created fault-injection profiles and JUnit tests across successive stages from refund to financial notification, verifying business error handling, system retries, status-update failures, and recovery.

**DataAnnotation | AI Trainer — Software Engineering** Oct 2026 – Present
- Design challenging bug fix and feature development tasks to expose coding agent failures; develop reference implementations and automated test suites for implementation-based evaluations.
- Evaluate agent reasoning and generated solutions against task requirements, identifying subtle logic errors, flawed implementation decisions, missed edge cases, and unreliable tests.

## Projects

**Tempo** | ***React, TypeScript, Node.js, Express, MongoDB, Better Auth, Railway*** 2026–Present
Full-stack calendar focused on making scheduling faster and more intuitive through simpler interactions.
- Independently designed, developed, and deployed Tempo end to end, implementing account authentication and user-scoped event storage so users can access their own calendars across sessions and devices.
- Engineered a regex parser that outperforms Google and Apple Calendar across 3+ tested scheduling scenarios, recognizing multi-day and combined date-time ranges (with "now" endpoints) in one field.
- Devised a one-click time entry workflow with instant AM/PM toggling for faster event creation.

**Personal Portfolio** | ***React, TypeScript, HTML/CSS, Vite, GitHub Pages*** 2026
Interactive personal portfolio presented through a custom Mac-inspired, pixel-art browser interface.
- Created a portfolio featuring original pixel art, custom animations, and interactive React components.
- Reduced five large media assets by 94% (10.8 MB to 0.64 MB) through resizing and WebP conversion, significantly cutting initial load time from approximately 5–6 seconds to under 1 second.

**GirlMath** | ***Python, FastAPI, React Native, TypeScript, MongoDB, DigitalOcean, Gemini API*** 2026
AI powered journaling app supporting girls in STEM by tracking confidence, motivation, and learning trends.
- Led backend dev in a 36-hour hackathon, built FastAPI logic and REST endpoints for a 4 person team.
- Integrated the Gemini API to derive confidence, motivation, and learning trends from journal entries.
- Built frontend API client for backend communication and deployed the FastAPI service to DigitalOcean.

**Munchables** | ***Java, Swing, OkHttp, JSON, Spoonacular API, HTML/CSS, JUnit*** 2025
Four-person Java Swing social platform created with Clean Architecture and JUnit-tested domain logic.
- Designed a reusable Swing post component across all views, centralizing image rendering, likes, and interactions, & implemented persistence using Clean Architecture, decoupling domain logic from storage.

## Skills

- **Programming Languages:** Java, Python, TypeScript/JavaScript, SQL, C, HTML/CSS, Assembly
- **Technologies & Tools:** Spring Boot, PostgreSQL, AWS, React/React Native, Node.js, Agents/Codex, Express, MongoDB, FastAPI, Maven, JUnit, Git, Expo, Railway, DigitalOcean, Java Swing
- **Languages:** Proficient in English, French, and Chinese